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Master of Software Engineering – 180 credits

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1 Task 1

1 Introduction to ITS and Dublin Bus

The challenges encountered in the urban transport systems include increasing congestion, environmental issues, and increasing demand of the mobility. Intelligent Transportation Systems (ITS) are becoming a popular solution to these problems in cities (including Dublin) worldwide. Dublin Bus being the main mass transportation in the city has gradually adopted ITS technologies to improve efficiency, reliable services and customer experiences.

Intelligent Transportation System(ITS)

City transport networks the world over are undergoing major challenges of rising urbanization, increased demand of the use of the individual vehicle and the environmental issues. To overcome these challenges, Intelligent Transportation Systems (ITS) have been strategic to shift away as a solution, which focuses on the integration of transportation engineering, communication technologies and data-driven decision-making to develop efficient, safe and sustainable mobility system (Sussman, 2018).

In the broadest sense, ATS/ITS is a system that comprises an extensive set of applications including automated fare collection to high use fleet monitoring, traffic management and real-time passenger information systems. Theoretically, the ITS is highly focused on integration, interoperability, and dynamic adaptation aspects of the ITS as real-time data should be continuously captured and processed and implemented to maximize the efficiency of the transport system (Papageorgiou et al., 2020).

Dublin Bus

In Dublin, Ireland, the problems associated with public transport, or more specifically the lack thereof, have been issues of congestion, declining ridership and poor levels of service reliability. Dublin Bus, which runs approximately 1,000 buses in the greater Dublin area of operation, has been the main transportation mode of the people with over 120 million passengers per annum1. Nonetheless, there was an inefficiency in communication, the usage of obsolete manual control, and unreliable traffic patterns, which in many ways negatively affected the operations.

As part of its modernisation, Dublin Bus installed a variety of ITS applications that included Electronic Ticket Machines (ETMs) in 1989, Automatic Vehicle Location and Control (AVLC) systems in 2010 followed by Real-time passenger information (RTPI) in 2011. These innovations reflect the gradual scaling to a hypothetical mode of an ITS, which may lead to the enhancement not only of the efficiency of computerized service delivery but also communication.

This report provides a detailed analysis of the ITS implementation by Dublin Bus and supplements with a review of the ITS literature and case studies of the ITS implementation across the world and recommends the general direction that Dublin Bus should pursue in the future, especially the use of artificial intelligence (AI) and machine learning (ML).

2 Literature Review

2.1 Theoretical Principles of ITS Design

In its design, its approach is informed by system theory in which transport networks are considered dynamic and adaptive systems that must be continuously evaluated in pursuit of an optimal performance (Papageorgiou et al., 2020). The main theoretical principles are the following:

On the move data capture and feedback: Data feedback and capture of vehicles and infrastructure guarantee responsive interventions in real-time (Wang et al., 2018).

Interoperability and integration: ITS must provide communication between the subsystems (e.g. fare collection, fleet monitoring and passenger information) through common standards (Barceló, 2017).

Human-system interaction: ITS solutions are socio-technical, so the adoption of a solution must be accepted by operators, drivers and passengers (Molin et al., 2020).

Scalability and versatility: As the number of urban residents grows, ITS structures will be forced to accommodate growth and improvement of technology (Zhang et al., 2020).

CEN Transmodel and SIRI are theoretical standards of data exchange used in the context of the public transport industry cementing compatibility between various operators and hardware (Barceló, 2017). These models focus on systematised data transmission between the cars, dispatching agencies and passengers.

2.2 ITS in Improving Software Services and Communication

ITS contributes to software services and communications in the transport systems in the public sector in a big way. AVL systems help an operator track buses in real time, and are vital to scheduling, incident response, and real-time passenger information. This enhances communications between drivers and control centers to make operations shift to proactive regulation techniques, compared to reactive problem resolving (Wang et al., 2018).

- In the same vain, Real-Time Passenger Information (RTPI) systems provide the expected time of arrival via apps, websites and on bus-stops. According to Chen and Zahavi, (2019), RTPI improves the subjective waiting time of passengers and their satisfaction even though the actual service frequency does not change (Chen & Zahavi, 2019).
- Another ITS communication-enhancing innovation is the Electronic ticket Machines (ETMs) and smartcards. The systems mitigate fraud, eliminate the amount of cash handled by drivers and provide historical data sets that can be used to analyze demanding loads. Gkiotsalitis and Cats (2021) highlight that fare collection conducted with the use of software allows transport agencies to predict the demand and optimals (Gkiotsalitis & Cats, 2021).

The ITS is also beneficial to internal communication to any staff safety and operations. Calls made on trunked radio systems incorporated into ETMs provide secure voice communications initiated and controlled by the dispatcher and ensures the driver is up-to-date with information without compromising passenger security.

2.3 International Case Studies and Good Practices.

Scalable Intelligent Transport Systems are systems that have been implemented worldwide to make urban mobility more efficient, accessible and sustainable. There are global examples that follow how these cities have implemented ITS solutions to facilitate transport systems using advantage of real-time information, smart technology and policy coordination. Scalable ITS can offer good insights like:

- **London (UK):** The transport network serves as an example of advanced ITS application in London to maximize the effectiveness and experience of people using road transportation. The RTPI system is a system integrated with Oyster card smart payment system that allows the collection of travel pattern details and fare information. White and Morton (2018) note that such integration enables the transport authorities to undertake demand-responsive planning, including varying the bus frequency to passenger demand, and reducing high-peak period congestion.

  Traffic Signal Priority (TSP) is another important application of ITS in London, and it enables buses to demand a longer green light at some intersections. This minimizes dwell time and dwell delays and enhances the service reliability. The RTPI, along with TSP, has resulted in measurable increases in bus ridership, increased compliance with schedules in a timely manner, and optimum use of resources.

- **Singapore:** Intelligent Transport System (ITS) implementation in Singapore is well coordinated to the overall transport policy of the city-state and considerable alignment between technology and governance is observed. According to Phang and Toh (2019), Singapore implements ITS in the form of Electronic Road Pricing (ERP) a dynamic congestion pricing system at the national level that limits traffic during peak time; multimodal smartcard ticketing (e.g., EZ-Link) on buses, trains, and other transportation types; and predictive analytics that predicts the demand of transportation and optimizes traffic flows. The country-wide integration of Singapore transport system enables ITS to improve efficiency in operations and contribute to wider objectives like sustainability, emission mitigation and urban livability. It results in improved modal changes, improved passenger assignment, and improved fluent policy making (Phang & Toh, 2019).

- **Tokyo (Japan):** The ITS in Tokyo is also a demonstration of the benefits of interoperability and real-time data analytics in a highly-populated transport system located in an urban population. The Suica and Pasmo smartcards are entirely interoperable through buses, trains, and underground systems, making commuting convenient to the user. Yagi (2020) concluded that when RTPI is used together with data on smartcard usage, the operators can track passenger flows in real-time, redistribute passengers to avoid overcrowding, as well as optimize train and bus schedules according to demand trends. This high digitalization enhances efficiency,

reduced congestion, and easier and stress-free transportation process.

- **Stockholm (Sweden):** Stockholm is using open data standard (Transmodel, SIRI) to enhance the integration of multimodes. Barloco (2017) adds that open data enhances travel planning, transparency and decision-making that help government authorities and developers to simplify services and create passenger-facing applications that facilitate efficiency and accessibility between modes of transport (Barceló, 2017).

This set of international best practices show that technology is not the only factor to the success of ITS but rather alignment of policies, funding, and support within the institutions.

2.4 Challenges and Scopes of Adoption of ITS

Despite the immense benefits of Intelligent Transport Systems (ITS), there exist difficulties when it comes to the implementation of these systems. To plan and implement successfully, it is important to understand these barriers.

- **Capital expenditures:** High initial cost is also one of the main barriers to the adoption of ITS. Real-time Passenger Information (RTPI) displays, Automatic Vehicle Location (AVL) systems, and other related communication infrastructure are expensive to implement (Li et al., 2021). In addition to procurement, constant maintenance, software upgrades and employee training contribute to the costs of running. Cities with large bus systems may even have a difficult time making the first investment particularly when other transportation and social infrastructures are a priority.

- **Integration issues:** The success of ITS is dependent on the integrity of the integration among various systems, such as fleet management, ticketing, traffic lights, and passenger information systems. However, transport operators also have disconnected or legacy systems, and such systems also lower the interoperability level and limit the full potential of ITS (Barceló, 2017). Without standard data formats and protocols, real-time exchange of information is inefficient and shrinks responsiveness and predictability. The operators tend to work with disconnected systems which hinder interoperability.

- **Staff and union resistance:** Human factor is another barrier. To some extent, ITS technologies, particularly surveillance systems such as AVL and automated trackers, can be perceived as a surveillance tool by transport staff, which subsequently results in

opposition or conflict with unions (Molin et al., 2020).The resistance may be in the form of unwillingness to use new working processes, lower adherence to data collection practices, or even direct resistance to ITS programs. An effective change management, communication and staff engagement strategies are therefore needed to ensure successful adoption.

- **Technical limitations:** Technological limitations can be limiting in terms of ITS performance. In Dublin, AVL polling rates of 20 seconds produce a latency, which kills such real-time applications as Traffic Signal Priority (TSP). Lag in data collection minimises the dynamic response capability of the system to congestion or operational disturbances. Other constraints can be the bandwidth of the network, reliability of the sensor, and lack of integration with traffic control infrastructure which in turn affects the overall functionality of ITS implementations.
- **Framework and institutional alignment:** It is these difficulties that highlight the need to integrate ITS implementation with institutional, financial and technical structures. Cities require detailed policies and funding plans and technical requirements to make ITS solutions scalable, sustainable and interoperable. Removing these barriers during the early phase of the planning process increases the chances of success in the adoption and the maximization of investment in ITS infrastructure.

These issues drive home the need to match ITS implementation with institutional, financial and technical frameworks.

2.5 Emerging Trends in ITS Literature

New studies point to the disruptive power of Intelligent Transport Systems (ITS) through Artificial Intelligence (AI) and Machine Learning (ML). Zhang et al. (2020) also claim that AI-driven ITS can transition to the next level of analytics, reactive to anticipatory and prescriptive analytics to enable transport operators to predict congestion, optimise service schedules, and improve the system overall efficiency (Zhang et al., 2020). Applications include:

- **Reinforcement Learning for Dynamic Scheduling :** Wei et al. (2019) note that reinforcement learning algorithms have the potential to optimise the bus and train schedules in real-time, dependent on the change in traffic conditions and passenger demand trends (Wei et al., 2019).

- **Neural networks in the forecasting of demand :** Li et al. (2021) show that this particular task also has applications with neural networks to predict passenger demand and prevent unnecessary resource allocation and overcrowding (Li et al., 2021).

- **Predictive maintenance of vehicles by using IoT:** Chen et al. (2020) state that artificial intelligence models and IoT-based sensors can help predict the breaking of a vehicle before it occurs, increasing the reliability of a fleet and minimizing the number of interruptions (Chen et al., 2020).

These innovations are indicative of the transition to self-optimizable ITS ecosystems in which real-time data is not only exploited to respond directly and reactively to congestion, but also to proactively manage congestion or plan operations and maintenance strategies. Introduction of AI and ML will enable ITS to become smarter, dynamic and better in retaining sustainable urban mobility.

There are several common themes that can be identified in the ITS literature: (1) **interoperability and open standards** as the means of offering scalable multimodal services (Transmodel, SIRI); (2) **real-time data** capture and the minimized latency as the conditions of the active traffic control (AVL/RTPI); and (3) **the policy-technology alignment** to make deployments generate system-level benefits (Papageorgiou et al., 2020; Barceló, 2017). Empirical evidence indicates that RTPI and smart ticketing lead to higher passenger satisfaction and allow to analyze the demand, but the effect of operational performance depends on being integrated with fleet control and signal priority (Chen & Zahavi, 2019; Gkiotsalitis & Cats, 2021). Recent publications point to AI/ML as a step towards the future of (reactive) to prescriptive ITS in optimising schedules, predictive maintenance, and demand forecasting, but with these works also raising concerns about cost, quality of the data used and the ability to deploy it (Zhang et al., 2020; Li et al., 2021). In general, the literature recommends additional mixed-method assessments that will integrate theoretical models with quantified KPIs (on-time performance, headway adherence, delay per vehicle) to allow cities to compare their results, not only to implement technology. The case of Dublin illustrates these themes: Dublin has invested in the fundamental ITS elements, yet the literature indicates that the last step to closer integration of standards, latency and policy integration will be the key toward quantifiable improvement.

3 Overview of Dublin Bus ITS Implementation

The Dublin Bus has implemented its Intelligent Transport System (ITS) using the evolutionary strategy, in which the bus has been implementing technological solutions that can result in enhancement in operational efficacy, consumer experience, and information-based orientation in decision-making. The key elements of the ITS of Dublin Bus are:

- **Electronic Ticket Machines (ETM):** Dublin Bus uses ETMs as the basis of its ITS installed in 2005. These machines get connected to peripheral devices, e.g. smartcard readers, destiny scrolls through RS485 communication interfaces. The ETMs are very relevant in terms of fare collection, cash reconciliation and back-office data aggregation to help us know how many tickets we sold and automatically report on the Financial and Operational Planning. Human error is also reduced and administration simplified with the ETMs, as the fare administration is computerized.

- **Automatic Vehicle Location Control (AVLC):** AVLC system came to use in 2010 and it relies on GPS and odometer data to monitor the buses with 20 seconds polling time interval. The real-time location information is transmitted to a centralized control center, where they can constantly monitor fleet activities, schedule compliance, and service interruptions. AVLC information is important to assist in Real-Time Passenger Information (RTPI) and operational decision-making including rerouting buses and deployment optimization during high traffic times.

- **Real-Time Passenger Information (RTPI):** Introduced in 2011 RTPI provides bus arrival information to passengers displaying electronic bus-stop terminals as well as on the Internet. As the project conducted under the supervision of Dublin City Council, RTPI is based on the constant stream of data provided by Dublin Bus, which guarantees the timely provision of the correct information on the passengers. RTPI improves commuter satisfaction, perceived waiting times and informed travel decisions.

- **Systems of communication:** Dublin Bus has been able to enhance operational communication through the integration of trunked radio systems and ETMs, which allow dispatcher-controlled messages to be sent straight to buses. This integration increases driver safety, provides effective management of operational incidences, and minimises the use of field inspectors. The communication system allows the control center and the vehicles to rapidly coordinate their response in the event of a breakage or

emergence.

- **Back-office Systems:** The information generated by ETMs and AVLC is inputted into back-office systems in order to help in financial reconciliation, operational planning and PSO contract management. These systems allow Dublin Bus to scale performance measures, optimize schedules and routes, and make decisions using information on service improvements and resource allocation.

Outcomes and Limitations

A more judgemental approach reveals that the ITS implementations by Dublin Bus have been mixed. Many real-life gains were also brought about by the adoption of Electronic Ticket Machines (ETMs) and smartcard functionality, such as less cash handling, better revenue protection, and the availability of usable fare datasets to aid planning (Gkiotsalitis & Cats, 2021). RTPI positively transformed perceived waiting time and customer satisfaction, which is consistent with the evidence worldwide, but has a relatively small effect on operational punctuality when Automatic Vehicle Location (AVL) polling and signal control are kept to a minimum (Chen & Zahavi, 2019). Although AVL systems offer much-needed fleet visibility and back-office planning, the current 20-second polling frequency limits the utility of AVL systems to applications with latency requirements, including dynamic Traffic Signal Priority (TSP) or real-time re-dispatching. Theoretically, ancillary analytics exists but due to the absence of data quality, integration, and governance, insights are sometimes hard to operationalize (Papageorgiou et al., 2020). Moreover, the small scale TSP pilot, and new Vehicle-to-Everything (V2X) testing indicates technical feasibility, but has shown funding, roadside infrastructure and standards gaps and needs more, city-wide coverage. In order to move out of the deployment phase and into measurable impact, Dublin Bus needs to prioritize quantifiable KPIs on time performance, headway compliance, mean delay per route, and vehicle downtime and release after and before analysis of results based on Transport for Ireland data to rigorously test interventions.

4 Future Direction and Recommendation

Advanced strategies that can help Dublin Bus expand its operations, customer satisfaction, and sustainability include:

- **Deploy AI and ML in computer infrastructure:** AI and ML algorithms can help Dublin Bus to handle congestion, anticipate passenger demand, and optimise the resources. One example of how AI can be utilized to improve demand forecasting is the possibility of predicting the busiest time of the day along a certain route and to dynamically allocate buses to reduce congestion and queues. Similarly, ML can analyse past history of traffic and ridership so that route planning and scheduling are optimised. This not just improves efficiency in the services but also reduces the costs of operation since buses get deployed when and where they are needed (Zhang et al., 2020).

- **Implement predictive maintenance using the IoT:** The concept of predictive maintenance that Dublin Bus can use is to equip its buses with IoT sensors that will track the performance of the engine, brakes, tires, and fuel flow on buses in real-time. These sensors could also be used to detect when a part is getting worn out or could possibly malfunction, so the maintenance staff can be proactive. This will contribute towards the reduction of unexpected failures, service failures and the serviceable life of the fleet, and decreased service costs in the long run (Chen et al., 2020).

- **Adopt state-of-the-art TSP tools:** To accomplish the ability to assign precedence to traffic lights in real-time, Dublin Bus may need to install low-latency 5G network-based Vehicle-to-Infrastructure (V2I) communications solutions. The buses will have the ability to communicate with the lights to request permission to have a green light or even extend their time, when a bus is at the front of the lights. This assists in reducing delays caused by traffic jam, enhancing route compliance and improving the reliability of the service. The rapid TSP systems are also interoperable with other urban transport systems in the provision of intelligent and integrated city environments.

- **Expand MaaS integration:** A unified solution to MaaS involves integrating the other transport modes like buses, trams (Luas), rail (DART), and a shared mobility solution in a single digital platform. Through one application, customers are able to book trips across more than two modes, access information and purchase tickets. The creation of these kinds of synergies gives Dublin Bus an opportunity to encourage people to use public transport, reduce individual car usage, and contribute to a more sustainable urban transport network. This can also facilitate co-ordination of schedules between modes of transport, making the whole network more efficient.

- **Improve communication with the passengers:** They can be implemented using mobile applications to personalise Real-Time Passenger Information (RTPI) systems to send alerts to passengers with an approximate arrival time and recommended alternate routes. With the implementation of multimodal journey planning within the apps, a passenger can simply alternate between bus, trams, and trains based on prevailing circumstances. Communication can be enhanced and consequently, more individuals will be persuaded to use the transport since it is reliable.

These would make Dublin Bus a technologically and modern public transport operator following international best practices. By using AI and ML, internet-of-things, TSP, MaaS, and better communication with customers, Dublin Bus will have an opportunity to make their operations more efficient, sustainable, and customer-satisfied. The creation of Intelligent Transport Systems (ITS) environment will also help develop the network to achieve future transport needs in the city, as well as assist transport measures that are sustainable and affordable.

Task 2

5 Challenges and Benefits

The challenges and benefits are below:

5.1 Challenges:

Dublin bus has various issues with congestion management:

- Congestion leads to a daily loss of 1,700km or 0.8% of total service planned to be given.
- Timekeeping is also a problem, because of unpredictable traffic conditions.
- The security concerns such as attacks of drivers and damage of property put a strain on staff relations.

Although Dublin has some of the common ITS problems that most cities have, a comparative analysis with international best practice demonstrates that the city has structural factors that restrict its performance in ITS. Cities such as Singapore and Stockholm have demonstrated that concerted policy, the uniformity of funding, and open interoperability standards is as relevant to technology implementation as is technology deployment (Phang & Toh, 2019; Barceló, 2017). In Dublin, the decentralized governance, legacy back-office systems and extended AVL polling times (20 s) limit real-time responsiveness and are unable to fully take advantage of Traffic Signal Priority (TSP). Where London and Tokyo have integrated RTPI,

smart ticketing and signal priority, Dublin with its weaker institutional integration stories rollouts piecemeal and provide very limited multimodal coordination. Resistance to surveillance and operational change by union and staff further decreases the successful adoption in the absence of a strong change management (Molin et al., 2020). The net impact of the same is that Dublin is likely to have the right components (ETMs, AVLC, RTPI) deployed, but with no governance, data standards and low-latency infrastructure (V2I/5G, roadside units) to transform those components into system-level benefits. It will be necessary to address these underlying challenges, such as policy integration, open-data standards, low-latency communications procurement and stakeholder engagement, to translate the investments in the ITS in Dublin into measurable service improvements (Barceló, 2017; Phang & Toh, 2019).

5.2 Benefits of ITS:

There are many benefits of Intelligent Transportation System:

- VLC implies that it can be dispatched proactive which makes it less inefficient.
- TPI does enhance customer satisfaction and demand in terms of ridership (Chen & Zahavi, 2019).
- Smartcards and TMs facilitate the collection of revenue and minimize fraud (Gkiotsalitis & Cats, 2021).
- Evidence-based planning and PSO compliance are supported with the help of data.

6 Comparation with international standards

The implementation of the ITS in Dublin has a number of gaps when compared to the best global cities. **In London**, Traffic Signal Priority (TSP) with Automatic Vehicle Location (AVL) systems have been successfully used to help reduce bus delays and increase compliance with schedules. The slowness of Dublin is mostly due to technical issues like rough AVL polling cycles, which do not allow real time responsiveness. **In Singapore**, ITS has deep roots in policymaking in Singapore through Electronic Road Pricing (ERP) and Mobility-as-a-Service (MaaS), and supports a multimodal integration. In Dublin, on the other hand, the integration and coordination of the various modes of transport is not fully captured in policies (Phang & Toh, 2019). Meanwhile, **Stockholm** is interoperable with open standards such as SIRI and Transmodel that provide the ability to exchange data across transport systems. The ITS in Dublin experiences poor data management, inhibiting the success of

traffic management and bus prioritisation projects (Barceló, 2017).

In sum, it can be noted that, even though Dublin has gone ahead in ITS implementation, real-time data management, integration of policies, and adherence to open-standard interoperability has to be improved to meet international standards.

ITS Component	Dublin Bus Outcome	Benchmark from Literature	Gap / Comment
ETM / Smartcards	Less cash handling, better protection of revenues, generation of useful fare datasets	According to the research conducted in the world, smartcards enhance the forecast of demand and minimize fraud (Gkiotsalitis & Cats, 2021).	No set standard of Dublin; good effect noted.
RTPI	Increased customer satisfaction and perceived waiting time.	It has been demonstrated internationally that RTPI increases satisfaction but does not have much effect on punctuality (Chen & Zahavi, 2019).	Dublin is representative of the global trend: perception is much better than punctuality.
AVLC	Facilitated fleet monitoring and planning; 20 seconds polling.	Advanced systems operate with 5 or less polling in order to facilitate real-time TSP or re-dispatching (Papageorgiou et al., 2020)	Dublin is technically lagging behind; poor support of dynamic TSP or re-dispatching.
Back-office Analytics	Analytics ability is there but not operationalised.	Data quality, integration, and governance are critical for actionable insights (Papageorgiou et al., 2020).	Dublin finds it difficult to operationalise analytics.
Traffic Signal Priority (TSP) / V2X	Pilot tests; also technically feasible, but not at city scale.	The deployment of these infrastructure, standards, and funding are necessary to execute successful city-wide deployment (Skabardonis, 2017).	Dublin not scalable; there are funding, infrastructure and standards gaps.

7 Traffic Signal Priority (TSP)

Traffic Signal Priority (TSP) The arrangements to change the timing of traffic lights to provide priority at the intersection to buses (public transport). TSP will contribute to enhancing schedule compliance, reducing delays and overall bus network efficiency by boosting bus time at red lights (Skabardonis, 2017).Unlike conventional traffic signal control that applies a fixed or adaptive control method, which is based on the overall traffic movement, TSP is a dymanic.

Dublin's Context:

i. **Create AVL polling interval :** Automatic Vehicle Location (AVL) is a tracking system used at Dublin Bus to track its buses. Currently AVL polling period is 20 seconds at a very coarse level. This data capturing delay prevents the effect of TSP because signal modification requires real time information. In order to explain, a bus moving towards an intersection may not be detected using the required speed to shorten or lengthen a green light or a red light; this implies that they will miss the opportunity of being prioritised.

ii. **Pilot V2X initiative :** A pilot program called Vehicle-to-Everything (V2X) is currently underway near the St. James Hospital that involves buses and traffic infrastructure communicating with each other. This pilot illustrates the opportunities to enable more responsive TSP. However, it cannot be deployed on a large scale due to a lack of funds and infrastructure that is currently available. The entire implementation would involve roadside unit investment, network integration and vehicle communications.

iii. **Real TSP Pilot :** Traffic Signal Priority (TSP) was tried on Quality Bus Corridors (QBCs) in Dublin in 2012 to 2016 (Lucan and Malahide). Buses fitted with Automatic Vehicle Location (AVL) reported to Dublin city council SCATS system where they asked to be given priority at traffic lights. Pilot showed 812% decrease in path times in the peak hours and extra reliability and schedule compliance of the buses, but expansive implementation was restricted by funding needs and the necessity to renovate signal controllers (Skabardonis, 2017, Chen & Zahavi, 2019). Positive results were observed with similar pilots internationally: the iBus TSP pilot in London, with an integrated AVL-based TSP, minimized delays at intersections by up to 16 percent and also increased passenger satisfaction, and the Green Light for Buses project in Stockholm used dynamic priority and real-time traffic information that reduced average trip duration by 5-10 percent and improved operational efficiency and reduced emissions (Barceló, 2017).

Recommendations:

i. **Develop low latency V2I across the 5G networks:** V2I networks using 5G must make sure that busses are communicating in real-time with traffic lights, allowing the signal to be adjusted in real-time. This will reduce bus delays, improve the overall traffic flow, and support connecting in the future to other related vehicles and smart city initiatives.

ii. **Dynamic Prioritisation based on AI:** The AI algorithms can prioritize the buses based

on its schedule delays, the number of passengers, and their future traffic needs. Included in the implementation are AVL polling update, deployment of roadside units, training AI models using past data, piloting in high density areas like the St. James Hospital and scaling throughout the city to result in improved punctuality and network performance.

8 Recommendations for AI and ML Integration

Strategic application of AI and ML can significantly enhance the Intelligent Transport System (ITS) implemented in Dublin Bus to be proactive and customer-centric in their activities.

- **Dynamic Scheduling:** Bus headways can be optimised in real time with the help of AI and ML (or reinforcement learning algorithms). This reduces bus bunching where a bus arrives at the same stop as another, causing arrivals and departures to be unevenly spaced and creating delays to passengers. Real-time optimization of the creation of headways which decreases bunching might also be performed using reinforced learning (Wei et al., 2019).
- **Maintenance:** Bus sensor data may then be fed to Ml models to identify trends that there is a potential component failure. To reduce unscheduled breakdowns, service interruption, and maintenance costs, maintenance will be scheduled in advance by estimating when a bus part may require maintenance. The maintenance model (reactive) is converted into the preventive (intervene) model (Chen et al., 2020).
- **Passenger Demand Forecasting:** Neural networks and other machine learning techniques can be used to predict passenger demand through the analysis of historical ridership data, seasonal patterns and other external factors (e.g., events, weather). Forecasts in real-time will allow the Dublin Bus to better redistribute buses and staff to ensure the availability of the right number of buses during peak periods and the availability of fewer buses during off-peak periods (Li et al., 2021).
- **The AI in question would be used primarily in prioritizing:** AI can also be used to optimise prioritisation of buses at traffic lights between the interests of the passengers and the interests of the rest of the traffic. After analyzing the traffic situation, passengers, and bus delays in real time, AI algorithms could generate intelligent decisions to extend green lights, or change the signal timings and improve the overall

performance of the network without a drastic effect on other road users (Zhang et al., 2020).

Impact on ITS:

Using AI and ML in planning, maintenance, predicting demand, and prioritisation of the services, Dublin Bus can build a data-driven and progressive ITS. This system enhances the passenger experience with increased reliability, reduced operation cost through predictive maintenance and optimisation of the networks through smart management of the traffic movement and resource allocation.

9 Conclusion

The case study on Dublin Bus also illustrates the life-changing influence of ITS in the modernization of States operations in the sphere of public transport. With speculated adoption of ETMs and RTPI, AVLC, and communication systems, Dublin Bus successfully enhanced the effectiveness of its operations and client customer service. But some issues still exist, in particular, in congestion control, integration of modes and deployment of TSP.

It can be seen that as compared to international best practices, there is a need to increase the level of policy integration, open standards usage, and congestion management techniques. The combined use of AI and ML offers a chance to shift Dublin Bus ITS model towards the predictive one and merge Dublin Bus with the leaders in terms of sustainable urban mobility.

Dublin has a significant opportunity to transform to a smart and sustainable transport system with an overarching role of Dublin Bus in digitisation and integration of our transport network.

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